

APPROVED

MSc IT for the Energy Industry
School of Computing, Engineering and Technology

Version	5.0
Course Type	Postgraduate
Award Type	Master of Science - MSc
Awards	
No Course Award Assigned	
Code	0621
SCQF Level	Level 11
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2026 September as of (September 2026)

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100366	computer science	computer science	100%
		Total	100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

What the Course Involves	The general format of the first two semesters for full-time students comprises a mix of lectures and lab sessions. Students will have access to all materials on the University's VLE. In semester 3, students are engaged on a project module, and have a structured timetable as well as meetings with their project supervisor to discuss progress and offer support. Part-time students attend the same programmed sessions as the full-time students but over an extended duration corresponding to two 15-credit modules per semester. Full Time Students - For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (project, taken in May) - For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 3 (project, taken in January)
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	Part Time Students • For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (taken in September), Semester 4 (taken in January), Semester 5 (project, taken in May and spanning 2 semesters) • For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 4 (taken in January), Semester 3 (taken in September), Semester 5 (project, taken in January and spanning 2 semesters)
Course Collaborations	
No collaborations listed	
Introduction to the Course	The Energy Industry relies on sophisticated computer hardware and software for a wide range of business activities from exploration to sales. While some products may be bought "off the shelf" and installed in-house, others are developed/customised/installed by a number of IT professionals. Therefore, there is a demand for IT specialists with sufficient knowledge of the industry, i.e. with a good understanding of renewable energy as well as oil and gas energy. The MSc IT for the Energy Industry aims to produce IT practitioners with training in software development tools and techniques, IT service management and a good understanding of the Energy industry. The course will prepare graduates in technical disciplines for a career in the design and implementation of information technology solutions and service management in the Energy industry and its IT providers. The course provides an integrated and coherent programme of education that lays the foundations for continuing professional development of the computing practitioner. The technical content of the course is directed towards the study of state-of-the-art software development tools and techniques as well as broad knowledge of the Energy industry. The <i>Software Project Engineering</i> module and the MSc Project are of particular value to employers as evidence of the ability to produce high quality work of an advanced technical standard to deadline and to a high standard of presentation.
	The aims of the course are to: • Provide knowledge and understanding of both renewable energy and oil and gas energy. • Produce specialist practitioners in the specification, design and implementation and adaptation of information systems. • Develop specialists in IT service management. • Enable students to enhance their legal, social, ethical and professional abilities including communication, presentation, critical appraisal and team-working skills. • Provide students with the foundations for professional development. Aims of Each Stage of the Course The specific aims for each stage of the course are: Postgraduate Certificate To produce practitioners who will be able to: • demonstrate knowledge and understanding of current software development technologies; • demonstrate knowledge and understanding of the Energy industry and the role of IT within it; • apply technical skills to the specification, design and implementation of moderately complex information systems.

Educational Aims of the Course	Postgraduate Diploma To produce practitioners who, in addition to the aims above, will be able to: • demonstrate knowledge and understanding of current database development technologies; • critically appraise characteristics and capabilities of underlying hardware and system software which can influence the choice and performance of software technologies; • establish a project management framework and lay down effective procedures for monitoring progress and quality assurance of a large scale development project; • manage the delivery of computer services; • demonstrate knowledge and understanding of the engineering processes relevant to the Energy industry and the role of IT within it. Master of Science In addition to the above, MSc graduates will be able to: • analyse the requirements of any unstructured and substantial problem discerning functional, human and performance factors; critically appraise different design approaches and technical solutions; • plan, manage and execute substantial software development projects within the framework of an appropriate software lifecycle model or networking problem; • integrate different software and information design technologies; • effectively test and document any complex software solution.
Awarding Institution	Robert Gordon University
Course Duration	Full-Time Students Full-time 12 months with a September intake Full-time 16 months with a January intake Part-Time Students Part-time 28 months with a September intake Part-time 32 months with a January intake Depending on their mode of study, students can choose to undertake either a short-term (3 months) or long-term (12 months) placement. All placements require approval from the School prior to commencement. If a student goes on placement, the duration of the course will be extended by the duration of the placement.
Distinctive Features of the Course	The course offers a combination of key modules to provide the student with a broad understanding and knowledge of renewable energy, petroleum geoscience and oil and gas engineering as well as information systems development and IT service management. The course will include interaction with IT professionals in the Energy sector. Therefore, students are well placed for a career in IT in the Energy Industry, as well as being well placed to pursue a career in the broader IT sector. There is a 3 month short placement opportunity, or a 12 month long placement opportunity, to develop the student's professional and technical skills in a business environment.
	Students may elect to undertake a short placement or a year-long placement.

Placement/Professional Experience	The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who the student can contact about any academic or non-academic problems during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short placement. This normally comprises 3 months of continuous, full-time supervised work experience. The assessment for a short placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short placement will normally receive at least 1 video call with their Academic Tutor. As an alternative, students may undertake a year-long placement. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. Placements must be undertaken after the completion of all taught modules and begin before the start of the final project. A student may choose to either undertake a placement at the same time as their final project or delay the start of their final project until the placement has finished. The course leader must approve this decision. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 180 credits achievable from the other modules. Successful completion is not required for progression or graduation. The credits achieved for placement may not be used to compensate for any credits from the other modules.
Opportunities for Further Study	The course exposes students to research techniques, enabling them to advance into research and development at a higher level (PhD).
External and Internal Indicators of Quality and Standards	The course has been design following the Scottish Credit Qualifications Framework credit value system. The course team has followed the Quality Assurance Agency guidelines and the British Computer Society guidelines for Masters-level course design. The School recently undertook a review of all its existing courses under the University's Institutional-Led Subject Review. The outcome of the review was very positive. The School benefits from teaching-research linkages. A significant number of academic staff also undertake research. A vibrant research community exists within the School, providing a rich environment for advanced study and the training to continue to PhD study or highly-skilled industrial posts.

	Projects are supported by our strong research groups in Machine Learning and Data Visualisation, Artificial Intelligence, Usability and Accessibility, Security and Privacy and Computing Education and by our excellent connections to industry.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Applicants should have an Honours degree and be able to demonstrate technical ability. This restriction may be relaxed for applicants with significant, relevant industrial experience. Applicants whose first language is not English should have an IELTS of 6.0 overall, with no component lower than 5.5, or an equivalent.
Alternative Qualifications for Entry	Applicants with a lower degree classification and / or relevant work experience will be considered on a case by case basis.
Interview Procedures	Generally, an interview is not required for admission to the course.
Course-Specific Regulations	The project modules, including the 15-credit investigation (CMM512 Project Investigation or CMM540 Workplace Project Investigation) and the 45-credit project (CMM513 MSc Project) cannot be condoned as they are key to demonstrating essential skills in the course such as the ability to conduct a significant piece of independent work. BCS guidelines state that the 60 credit project should not be compensated. CMM538 Short Industrial Placement cannot be used to compensate for failed modules. All students, regardless of mode and start date, will typically undertake resits in the summer exam period. Exceptionally students who start in January may be offered a resit opportunity after completing two semesters, in April of the following year.

Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: • To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.

Course-Specific Learning Experience

The MSc IT for the Energy Industry offers the students on the course practical IT experience together with knowledge and understanding of the industry. The course has a vocational focus where students must demonstrate the application of IT skills in an Energy Industry context.

The breadth and depth of the course demands a rapid pace of delivery using a variety of learning strategies including both traditional delivery methods and student-centred learning. Different modules may be delivered differently, depending on the nature of the subject, its learning outcomes and its role in the overall course aims and objectives as follows:

Core Technological Units

These modules are delivered through lectures, tutorials and, where appropriate, supervised laboratory sessions and supplemented, where appropriate, by student-centred activities and materials. The core technological modules are divided into two main disciplines:

- Energy related: modules ENM503 *Introduction to Renewable Energy* and ENM502 *Introduction to Oil and Gas Engineering* cover the energy industry in some detail.
- IT related: modules CMM024 *Object Oriented Programming*, CMM022 *Data Management*, CMM020 *Data Visualisation and Analysis* and CMM007 *Intranet Systems Development* cover a wide range of technical IT topics.

All technical modules involve practical sessions where students are expected to progress through a series of graded exercises that are designed to foster their analytical skills and to illustrate the synthesis of fundamental knowledge and problem solving techniques in a structured problem domain.

Professional Skills Module

Technical report writing skills are covered in all modules. Organisation and management skills are covered in the *IT Infrastructure and Service Management* unit CMM008. The *Software Project Engineering* module CMM004 and the Project modules CMM512 or CMM540 and CMM513 include presentation skills as well as project planning and project management. CMM004 also includes team-working and ethical, professional, social and legal issues are considered in CMM512 or CMM540 and CMM513 where the student undertakes an IT project, working through all stages from problem analysis and requirements specification to a well-documented software solution.

The course addresses project planning skills that are vocationally relevant and that also directly underpin the planning and management of the MSc Project. Student feedback in this course takes the form of critical self-appraisal, peer appraisal, and commentary from academic staff. Through directed reading and by listening to invited external speakers, students are also encouraged to develop a broader perspective of Information Technology and to sharpen their awareness of the need for professional standards and ethics in the workplace.

Integrative Course Modules

During the MSc project (CMM512 or CMM540 and CMM513) students combine the technical and professional skills learnt in previous modules in order to develop a substantial piece of software. Thus, students start by analysing project requirements and conducting a thorough critical appraisal of the problem domain and the techniques and tools which can be used in the solution. The students must also analyse the legal and ethical implications of their project at the start and, if appropriate, at later states in the project. Once a solution is implemented, a substantial evaluation must be conducted and the whole project must be documented. An oral presentation and demonstration is given at the end of the project by all students.

Research Skills

The course framework underpins the MSc Project in a number of ways:

- PgC / PgD taught modules provide support in specific subject areas,
- the Software Project Engineering module (CMM004) is designed to foster understanding and competence in project management and planning issues.

During the project investigation (CMM512 or CMM540) students gain both theoretical (via lectures) and practical experience of the research skills required in order to conduct a successful and critical appraisal of the literature, including emerging techniques and technologies, with respect to their project topic.

	All students who are about to embark on the MSc Project are required to attend a series of lectures, both at the Project Investigation phase and at the Project phase. The lectures cover all aspects of research, from the initial stages of objective definition, through information finding, design, evaluation, report writing and presentation skills. In addition, successful project investigations are discussed. Students also attend an information retrieval workshop presented by staff from the Library Service which is intended to familiarise students with available information sources (journals, monographs and on-line resources) and to inform them of systematic and effective methods to review current literature in the field of computing. The Workshop is not assessed directly but evidence of independent research and a review of the project context are elements of the overall assessment of the MSc Project. <u>Extracurricular Activities and Transferrable Skills</u> Students are encouraged to participate in several school organised events and student societies: • The school organises <i>Hackathon</i> events where students work over a 24 hour period solving unseen real-world coding and development problems in teams; • The School organises a <i>Computing Show</i> where students can demonstrate project, coursework or personal projects to industry visitors and staff and potential employers; • The School hosts regular BCS talks which students are encouraged to attend; • Students have the opportunity to attend coding and cyber security competitions and challenges; • Students from the School run a <i>Computing Society</i> that organises talks, events and social events.
	The course curriculum is designed to enable the student to acquire and develop their subject knowledge and understanding, intellectual and thinking ability and practical and key transferable skills at Masters level corresponding to the Scottish Credit and Qualifications Framework. As the students progress through the course they are encouraged to expand their understanding and critical appreciation of advanced computing, and develop skills in programming through techniques and tools with the application of this knowledge. The overall philosophy and strategy proposed for the course is to provide a sound knowledge and understanding of the IT in the Energy sector. The vocationally orientated nature of the course requires the student to achieve general and specific skills, which are addressed through a combination of teaching and learning methods, including lectures, lab sessions, tutorials, seminars, student centred learning and private study. Directed guest lectures emphasise the vocational nature of the course and further contextualise the technical content of the course within the energy industry. The course induction programme, with input from the library and the careers service, introduces the student to a variety of services via lecture, workshops or informal discussions. It also covers life in Aberdeen, School and University procedures, extenuating circumstances, academic misconduct and entrepreneurship. A handbook on basic study skills is provided to all students. Learning skills are developed through tutorials, directed reading and projects. Tutorials take the form of graded problem assignments e.g. in laboratory sessions; case studies. The theoretical underpinning of subject matter is provided by lectures. Regular meetings with project supervisors in MSc project modules supports the student in their investigative and reporting work as well as in the planning design, development and evaluation of an IT system. The broad nature of the course including common and specialist elements necessitates the use of a broad range of teaching techniques. These include lectures, seminars, tutorials, computer laboratories, open and distance learning and group work on a large project. <u>Lecture</u> This is usually a formal discourse for the purposes of dissemination of information, the demonstration of techniques and the discussion of supporting ideas and consequences. The lecture is supported by a full range of dissemination equipment including whiteboard, video and computer projection facilities where appropriate. Although this type of presentation is suitable for a one-sided discourse ample opportunity exists for questions, interaction and discussion.

Teaching and Learning Strategy	<u>Seminar and Tutorials</u> In the School of Computing, these two titles are used for a wide range of activities, each suited to the particular subject. On the one hand, some tutorials will consist of the staff supporting students engaged in problem solving. On the other hand, some tutorials may involve group exercises or discussions. Generally, this type of teaching is used to support the lecture, clarify the material and experiment with the techniques and skills required. The School of Computing runs regular seminars which cover a wide range of IT topics from a research and/or industrial point of view. Some of the seminars are given by School's staff whereas others are given by external (industrial or academic) speakers, including experienced IT professionals currently working in the Energy Industry. Students are also encouraged to join the RGU student chapter of the Society of Petroleum Engineers who run seminars and invite industrial speakers. <u>Laboratory</u> The nature of the computing elements of all courses requires students to gain practical skills in the use of a personal computer or workstation. This activity takes place in one of the School's computer laboratories and consists of the student, supported by a staff member, practicing skills in the use of sophisticated software applications and including software development and systems analysis and design tools. The Energy modules are taught by staff at the School of Engineering. <u>Project</u> The MSc Project serves the primary purpose of integrating technological, professional, social, ethical and research strands, which are developed in the preceding PgD stage, and does so in the context of a substantial research or information systems development project. The project typically involves the development and evaluation of an IT solution to a problem, which occurs within a relatively unstructured domain related to the energy sector. The problem is original to the student and its solution therefore requires the innovative application of knowledge and techniques, either studied in the previous PgD stage or acquired through independent research of recent and relevant literature. The project is spread over two modules CMM512 or CMM540 and CMM513, where the first module is a choice between a School-based academic investigation via CMM512, or a work-based placement and investigation via CMM540, both covering an initial investigation into the subject area of the project, and the clarification of a problem to be solved, and the second module CMM513 covers the design and implementation of a solution to this problem. The MSc Project provides a vehicle for integrating specialist knowledge with analytic, problem solving, managerial and communication skills. The first (investigation) modules CMM512 or CMM540 exercises and evidences skills in information gathering, critical appraisal and analysis and communication. The second module requires, in addition to problem solving, evaluation and managerial skills which are exercised and evidenced through the execution and outcome of the project, which include a project plan, dissertation, final oral presentation and project 'viva' (demonstration).
	A range of assessment methods is used, with particular emphasis on practical assignments (take away courseworks or assessed laboratory work), including design exercises (ranging from design studies to fully implemented solutions), comparative studies, investigative reports and unseen written examinations. Technical reports documenting software design and/or development assignments; structured project documents to manage project conduct, establish quality guidelines, archive project deliverables; use of WWW resources to present interim MSc Project progress; live presentation of MSc Project outcomes.

Assessment

Assessment of use of IT tools is included in all coursework assignments (i.e. use of tools to plan, aid design, collate results and document coursework).

Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions.

Assessment of proficiency in the use of collaborative IT tools for team working is required for the group-based software engineering project CMM004.

Assessment of interaction between student and project supervisor in conduct and defence of MSc Project.

Assessment by coursework deliverables is also used. Students submit technical report content which reflects on problem identification, alternative solution strategies, evaluation of solution performance.

Written examinations will be used to assess the students' understanding of technical modules.

Coursework deliverables will be used to assess problem identification, problem analysis with evaluation of alternative solution strategies, solution design, evaluation of solution performance, documentation. In some cases, courseworks will take the form of an assessed lab, where students undertake a coursework in a computer laboratory under exam conditions. During assessed labs, students normally have access to a range of reference materials.

The following sections detail the teaching and assessment of various skills including: (i) knowledge and understanding skills; (ii) practice: knowledge and understanding; (iii) generic cognitive skills; (iv) communication, ICT and numeracy skills and; (v) autonomy, accountability and working with others.

Knowledge and understanding skills

A range of assessment methods are used, with particular emphasis on coursework assignments, including design exercises (ranging from design studies to fully implemented solutions), comparative studies, investigative reports and unseen written examinations.

Practice: knowledge and understanding

Practical skills are assessed throughout via coursework assignments which involve the use of a variety of programming languages, IT and statistical tools and other computing tools and packages. Evaluation skills are key to demonstrating knowledge and understanding of information technology practice. A variety of practical skills are assessed in the MSc Project (CMM513) which involves the demonstration of the practical skills required to complete a substantial practical computing task.

Generic cognitive skills

Assessment generally involves the production of technical report content which reflects on problem identification, critical appraisal of alternative solution strategies, evaluation of solution performance. The assessment of generic cognitive skills is particularly evident in the two MSc Project modules (CMM512 MSc Project Investigation or CMM540 Workplace project Investigation and CMM513 MSc Project).

Communication, ICT and numeracy skills

Communication: Technical reports documenting computing/IT assignments; structured project documents to manage project conduct, establish quality guidelines, archive project deliverables; use of web resources to present interim MSc Project progress; live presentation of MSc Project outcomes. Team communication tasks are implicitly assessed in group work tasks which involve both oral/written communication between team members and well as document sharing.

ICT: The use of IT tools in all coursework assignments (i.e. use of tools to plan, aid design, collate results and document coursework). Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions including development of support materials for the project oral presentation.

Numeracy skills: these skills are essential for the successful completion of most course modules and are implicitly assessed in courseworks and unseen written examinations. Numeracy skills are particularly key to data analysis tasks and the evaluation of project results.

Autonomy, accountability and working with others

These skills are assessed implicitly in all modules. Explicit assessment of these skills is evident in the conduct mark of the project, which reflects the students interaction with their supervisor and includes demonstration of own-initiative and ability to work independently. Team-working skills are mainly assessed in the Software Project Engineering module (CMM004) which includes group work for the design and implementation of a software solution.

	It is important to note that all grades indicated to students during the course of the academic session are provisional until the deliberations of the Assessment Board for the Course. Provisional grades are presented to this Board, which consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each Module undertaken by them. This is subsequently confirmed to each student in an electronic transcript which may be accessed on the Student Portal.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	The School's computing facilities comprise 9 labs in total housed with over 335 state-of-the-art PCs and 30 Macs, each updated over a 3 year cycle to ensure they deliver high performance and support the latest software. School facilities include the School's Mac Suite, a soundproof booth used for podcasting, a Green Screen Suite used in multimedia video production and a high performance computer cluster used to create computer generated imagery. The School offers 100% wireless network coverage for laptops and other handheld devices. There's also a comprehensive range of software on-site - some developed by staff to support coursework and making wide use of open source and public domain technologies. Established relationships with major software suppliers mean access to the latest software. Access to the School's labs is by electronic key fob at any time during University opening hours. Students have access to the University's wireless network for laptops and other handheld devices. All students are given access to the Student Handbook, which contains a wide variety of information about the School and University. Students will be supported through the School's integration of the institutional Team Around a Student (TAAS) approach, ensuring that every student has a point of contact who can provide advice, triage queries, and connect them with

	the most appropriate support and resources where issues may be affecting their studies or wider student experience. The weekly induction sessions during the first semester of study cover a wide variety of topics such as library facilities, extenuating circumstances, exam preparation, careers services, placement opportunities, etc. All students are assigned a supervisor for their project work. In addition, they receive extensive guidelines on how to undertake a project. The School believes the academic success of the students is founded in the desire to develop and maintain close links with industry and to actively seek state-of the-art technologies and integrate these into course modules.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	The School of Computing is committed to the highest standards of quality and best pedagogy and research within an environment which values all associated stakeholders. Formal feedback is gathered from Student Questionnaires which are issued on completion of the PgDip stage of the course. It offers the students the opportunity to comment in detail on all aspects of the course as well as the general environment, support services and facilities. Responses to, and actions resulting from, these comments are addressed fully in an Annual Course/Programme Appraisal and widely discussed by the Postgraduate Management Team. The Programme Team also runs a series of informal focus groups in order to gather both positive and negative comments on the course. Where problems are highlighted, these are discussed fully and potential solutions are considered. In addition, the students have an opportunity to discuss with their Course and/or Programme Leader any course or management issues at any time. Less formally, the students have an opportunity to discuss with their Course Leader any course or management issues at weekly Academic Support sessions.
	The University and RGU:Union work in partnership to create an environment which stimulates:

University Student Feedback Mechanisms

- the participation of students, by empowering them to proactively provide views and opinions;
- a supportive learning community where students and staff engage in meaningful dialogue; and
- the engagement of students in the design of solutions and enhancements.

This approach is underpinned by the *Student Partnership Agreement* which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: [Student Representation & Partnership | RGU](#) .

There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience.

Mechanisms through which students are engaged and supported include:

- participation in student evaluation questionnaires and where relevant the *National Student Survey (NSS)* ;
- staff/student liaison arrangements at course/programme level;
- support for Student Representatives;
- student representation on Institution-Led Subject Review Panels and Validations;
- student representation on key University committees.

Course-Specific Student Feedback Mechanisms

A single Staff/ Student Liaison Committee exists for all postgraduate computing courses. The Committee comprises the Head of School, Course Leaders, the Programme Leader, the Student Officer, teaching staff and class representatives from each course intake and meets at least once per semester to exchange information and discuss issues.

The Student Partnership group also meets regularly and it is open to all students who want to join, offering a forum for students to comment on positive and negative aspects of their student experience.

The course team also runs a series of informal focus groups in order to gather both positive and negative comments on the course. Where problems are highlighted, these are discussed fully and potential solutions are considered. In addition, the students have an opportunity to discuss with their Course and/or Programme Leader any course or management issues during sessions in the Computer Academic Support Centre, or by arranging an individual appointment. Class representatives are also used by students to communicate issues or requests to the course team.

Awards**Stage 1**

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Postgraduate Certificate - PgCert	IT for the Energy Industry	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Diploma - PgDip	IT for the Energy Industry	Aberdeen, United Kingdom	Exit Award		120
Master of Science - MSc	IT for the Energy Industry	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Capable of demonstrating knowledge and understanding of the Energy Industry.	SC1	SCQF 11
LLO2	Able to align IT services with the needs of the Energy industry and its IT providers	SC1	SCQF 11
LLO3	Proficient in the use and customisation of modern software tools and environments.	SC1	SCQF 11
LLO4	Able to integrate the use of these IT tools and techniques in the framework of an appropriate development methodology to develop high quality software solutions.	SC1	SCQF 11
LLO5	Current database technologies.	SC1	SCQF 11
LLO6	The capabilities of modern software development methods and tools.	SC1	SCQF 11
LLO7	The analysis and specification of the information storage and processing needs of industrial and commercial applications.	SC1	SCQF 11
LLO8	How to plan and monitor the conduct of the development process with a view to ensuring delivery of a quality-assured product.	SC1	SCQF 11
LLO9	The frameworks and best practice in IT service delivery.	SC1	SCQF 11
LLO10	Appropriate research methodologies and detailed understanding of the particular area in which the Masters project is carried out.	SC1	SCQF 11
LLO11	Demonstrate the skills required to produce structured and coherent written reports to document coursework results and the investigation and survey of a problem domain.	SC4	SCQF 11
LLO12	Demonstrate the skills required to document research strategy and design decisions and to record research or test strategies and results throughout the various stages of a major research /information systems development project.	SC4	SCQF 11
LLO13	Demonstrate the skills required to record key progress made on the project on a weekly basis.	SC4	SCQF 11
LLO14	Demonstrate the skills required to share software and documentation between members of a team using a collaborative tool.	SC4	SCQF 11
LLO15	Demonstrate the skills required to deliver a clear, well structured, oral presentation to an audience with the aid of presentation software.	SC4	SCQF 11
LLO16	Demonstrate the skills required to defend the work undertaken during the project in an oral question-answer session.	SC4	SCQF 11
LLO17	Demonstrate the skills required to produce a clear and coherent thesis which documents the process and end result of a major research/ information systems development exercise.	SC2	SCQF 11
LLO18	Demonstrate the skills required to use software tools for the preparation of technical documentation and presentation related to the course.	SC2	SCQF 11
LLO19	Demonstrate the skills required to effectively use online research and learning resources.	SC2	SCQF 11
LLO20	Demonstrate the skills required to use software tools and packages to undertake practical work for courseworks, for collaborative tasks and for the project.	SC2	SCQF 11
LLO21	Demonstrate the skills required to work independently, managing his own learning and utilising the resources and study techniques appropriate to the course-specific modules.	SC3	SCQF 11
LLO22	Demonstrate the skills required to work effectively under the guidance of a project supervisor and as part of a collaborative team	SC3	SCQF 11
LLO23	Demonstrate the skills required to conduct a full requirements analysis for an information systems problem.	SC3	SCQF 11
LLO24	Demonstrate the skills required to apply appropriate management skills to the effective provision of IT services.	SC3	SCQF 11

LLO25	Demonstrate the skills required to propose, using appropriate documentation and notation, several solutions to an information systems need in the context of the Energy industry.	SC3	SCQF 11
LLO26	Demonstrate the skills required to compare and contrast proposed solutions to an information systems need, identifying key distinguishing advantages and/or disadvantages of the various solutions.	SC3	SCQF 11
LLO27	Demonstrate the skills required to select and apply appropriate testing and validation techniques to an information system solution.	SC3	SCQF 11
LLO28	Demonstrate the skills required to evaluate an information system solution with reference to the initial aims and objectives.	SC3	SCQF 11
LLO29	Demonstrate the skills required to select and conduct the Masters project, applying an analytic, rigorous and critical approach to identifying the project goals, proposing and evaluating different solution strategies and generating a useful and effective output.	SC3	SCQF 11
LLO30	Demonstrate the skills required to work, as part of a team.	SC5	SCQF 11
LLO31	Demonstrate the skills required to effectively manage the provision of IT services.	SC5	SCQF 11
LLO32	Demonstrate the skills required to design, develop, test, evaluate and document an appropriate information system solution to a challenging problem, using suitable analytical techniques in the process.	SC5	SCQF 11
LLO33	Demonstrate the skills required to investigate and, critically appraise specialist areas of research.	SC5	SCQF 11
LLO34	Demonstrate the skills required to use appropriate tools and techniques to analyse complex data.	SC5	SCQF 11
LLO35	Demonstrate the skills required to integrate appropriate critical appraisal and analysis into all aspects of the work.	SC5	SCQF 11
LLO36	Demonstrate the skills required to: consider and assess ethical and legal issues with respect to own work and the management of others.	SC5	SCQF 11
LLO37	Demonstrate the skills required to conduct an ethical and legal assessment of proposed work.	SC5	SCQF 11
LLO38	Demonstrate the skills required to clearly communicate complex technical information orally and through written and visual work, using appropriate terminology and documentation conventions.	SC5	SCQF 11
LLO39	Demonstrate the skills required to apply a range of learning techniques, including learning from others.	SC5	SCQF 11
LLO40	Demonstrate the skills required to work independently, managing the student's own workload, establishing contingencies into the planning of tasks.	SC5	SCQF 11
LLO41	Demonstrate the skills required to thoroughly evaluate work in an objective manner.	SC5	SCQF 11
LLO42	Demonstrate the skills required to collaborate with others to develop an IT solution to a problem.	SC5	SCQF 11
LLO43	Demonstrate the skills required to analyse the legal and ethical issues related to their work.	SC5	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7

SC2	Practice: Applied knowledge, skills and understanding	SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A: Non-placement Route B: Short Placement Route C: Long Placement
Part Time On Campus	Route A: Non-placement Route B: Short Placement Route C: Long Placement

Stage 1 / Semester 1**Full Time On Campus**

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Part Time On Campus

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CMM022	Data Management		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11
Full Time On Campus	CMM028	IT Infrastructure and Administration			Yes	1.0	15	Level 11
Full Time On Campus	CMM024	Object-oriented Programming		Jean-Claude Golovine (jrrg5708)	Yes	1.0	15	Level 11
Part Time On Campus	CMM024	Object-oriented Programming		Jean-Claude Golovine (jrrg5708)	Yes	1.0	15	Level 11
Full Time On Campus	ENM275	Renewable Energy Technologies		Dallia Ali (da4630)	Yes	2.0	15	Level 11
Part								

Time On Campus	ENM275	Renewable Energy Technologies		Dallia Ali (da4630)	Yes	2.0	15	Level 11
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Stage 1 / Semester 2**Full Time On Campus**

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Part Time On Campus

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CMM007	Intranet Systems Development			Yes	1.0	15	Level 11
Part Time On Campus	CMM007	Intranet Systems Development			Yes	1.0	15	Level 11
Full Time On Campus	ENM502	Introduction to Oil and Gas Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 10
Part Time On Campus	ENM502	Introduction to Oil and Gas Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 10
Full Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
Full Time On Campus	CMM004	Software Project Engineering		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11

Stage 1 / Semester 3**Full Time On Campus**

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Part Time On Campus

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time On Campus	CMM022	Data Management		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11
Part Time On Campus	CMM028	IT Infrastructure and Administration			Yes	1.0	15	Level 11
Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Stage 1 / Semester 4**Part Time On Campus**

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part								

Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
Part Time On Campus	CMM004	Software Project Engineering		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11

Stage 1 / Semester 5**Part Time On Campus**

Route A: Non-placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time On Campus	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Part Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Full Time On Campus	CMM022	Data Management		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CMM028	IT Infrastructure and Administration			Yes	1.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CMM024	Object-oriented Programming		Jean-Claude Golovine (jrrg5708)	Yes	1.0	15	Level 11
1	Semester 1	Core	Part Time On	CMM024	Object-oriented Programming		Jean-Claude Golovine	Yes	1.0	15	Level 11

			Campus			(jrrg5708)					
1	Semester 1	Core	Full Time On Campus	ENM275	Renewable Energy Technologies		Dallia Ali (da4630)	Yes	2.0	15	Level 11
1	Semester 1	Core	Part Time On Campus	ENM275	Renewable Energy Technologies		Dallia Ali (da4630)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CMM007	Intranet Systems Development			Yes	1.0	15	Level 11
1	Semester 2	Core	Part Time On Campus	CMM007	Intranet Systems Development			Yes	1.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM502	Introduction to Oil and Gas Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 10
1	Semester 2	Core	Part Time On Campus	ENM502	Introduction to Oil and Gas Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 10
1	Semester 2	Core	Full Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CMM004	Software Project Engineering		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11
1	Semester 3	Core	Part Time On Campus	CMM022	Data Management		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11
			Part								

1	Semester 3	Core	Time On Campus	CMM028	IT Infrastructure and Administration			Yes	1.0	15	Level 11
1	Semester 3	Core	Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11
1	Semester 4	Core	Part Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time On Campus	CMM004	Software Project Engineering		Kit-ying Hui (cmskhu)	Yes	1.0	15	Level 11
1	Semester 5	Core	Part Time On Campus	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 5	Elective	Part Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 5	Elective	Part Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Aberdeen, United Kingdom	On-Campus	Part Time	
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Sources of Further Information

No sources of further information recorded
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